# ESSAM MOUSSA MAHMOUD

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## Professional Summary

Full-Stack Developer with hands-on MERN experience (MongoDB, Express.js, React.js, Node.js) building role-based, real-time, and API-driven web applications across 4 production-style projects. Implements secure authentication (JWT, OAuth, bcrypt), concurrency-safe operations using MongoDB transactions, and analytics with MongoDB aggregation pipelines. Comfortable owning a feature end-to-end — schema design, REST API, and responsive React UI — from requirements through deployment. Recently expanded into PHP, Laravel, and MySQL, adding a relational-database and MVC-framework perspective to a JavaScript-first skill set.

## Technical Skills

**Languages:** JavaScript (ES6+), TypeScript, PHP, HTML5, CSS3, SQL, Python, C++
**Frontend:** React.js, Redux, React Query, Tailwind CSS, Material UI, Shadcn/UI, Responsive Web Design
**Backend:** Node.js, Express.js, Laravel, REST API Design, JWT Authentication, OAuth, Socket.IO
**Databases:** MongoDB, Mongoose, MySQL, Transactions, Aggregation Pipelines, Database & Schema Design
**Tools & Practices:** Git, GitHub, Postman, Vercel Deployment, Agile/Scrum Fundamentals

## Experience

**Full-Stack Development Intern — TechTrek** *Jul 2026 – Sep 2026*

- Completed a 2-month full-stack curriculum (React, Next.js, TypeScript, Node.js, Express, MongoDB), progressing from frontend fundamentals through backend architecture, database integration, authentication, and deployment.
- Delivered 2 full-stack projects during the internship, including a capstone project spanning both frontend and backend development.

**Full-Stack Development Intern — Information Technology Institute (ITI)** *Aug 2026 – Sep 2026*

- Completed a backend training internship covering PHP, the Laravel framework, and MySQL, finishing every lab and project assigned throughout the program.
- Practiced core Laravel concepts — routing, controllers, and Eloquent ORM — alongside relational schema design and querying in MySQL through hands-on lab work and project assignments.

**Full-Stack Web Development Training — Digital Egypt Pioneers Initiative (DEPI)** *Nov 2025 – Jul 2026*

- Shipped full-stack MERN applications on sprint-style milestones within a structured, project-based curriculum, collaborating with peers in a Git/GitHub team workflow with code review.
- Designed and implemented REST APIs against both SQL and NoSQL databases, delivering end-to-end CRUD and business-logic features under fixed deadlines.
- Architected reusable component libraries and implemented client-side routing and state management across multiple production-style applications.
- Debugged and resolved application-logic defects under deadline pressure to deliver demo-ready features for milestone reviews.

## Projects

**EasyCare — Healthcare Appointment & Patient Management Platform** *2026*
*React.js · Node.js · Express.js · MongoDB · Mongoose · Socket.IO · JWT · bcrypt · Leaflet · Recharts*

- Architected and shipped a full-stack MERN healthcare platform supporting 3 role-scoped user types (patient, doctor, admin), with ownership-based authorization enforced at the API layer to prevent unauthorized cross-user access.

- Built a concurrency-safe appointment booking engine using MongoDB multi-document transactions, eliminating race-condition double-bookings under simultaneous requests.
- Implemented secure multi-method authentication (JWT, bcrypt password hashing, Google OAuth) with role-based authorization middleware protecting all backend endpoints.
- Delivered real-time patient-doctor messaging via Socket.IO and an admin analytics dashboard (Recharts) powered by custom MongoDB aggregation pipelines for reporting and review moderation.

**Wedding Halls Booking Platform** *2026*

*React.js · Node.js · Express.js · MongoDB · REST API*

- Delivered a full-stack MERN venue discovery and booking platform as a freelance engagement for an independent client, from requirements gathering through a REST API-driven Express/Node.js backend.
- Designed MongoDB schemas for venues, availability windows, and booking records, exposing RESTful endpoints consumed by the React frontend.
- Developed a responsive React UI for venue discovery, including detail pages with pricing, availability, and image galleries.
- Executed a full visual/UX redesign based on direct client feedback, elevating the platform from a functional MVP to a polished, production-grade product ready for launch.

**Library Management System** *2025 – 2026*

*React.js · Redux · React Query · Tailwind CSS · Shadcn/UI · Recharts · Zod*

- Engineered a library management system handling book inventory, user accounts, and borrow/return workflows through a reusable, component-driven architecture.
- Implemented schema-validated forms with Zod and a secure authentication flow with sanitized input handling and structured error states.
- Separated global UI state (Redux) from server state (React Query) to enable real-time borrow/return updates without unnecessary re-fetching, improving perceived responsiveness.
- Structured the codebase around reusable, typed components to support long-term maintainability.

**Admin Dashboard** *2025*

*React.js · Material UI · Nivo Charts · FullCalendar · Formik · Yup*

- Designed a responsive, modular admin dashboard using React and Material UI, prioritizing component reusability across views.
- Implemented interactive data visualizations using Nivo Charts to surface key operational metrics at a glance.
- Integrated calendar-based scheduling and event management with FullCalendar for streamlined operational workflows.
- Built validated, reusable form components with Formik and Yup, reducing invalid-input errors through client-side validation.

## Education

**B.Sc. Computer and Data Science (In Progress) — Alexandria University** *Oct 2023 – Expected Jun 2027*

## Certifications

- Full-Stack Web Development — Digital Egypt Pioneers Initiative (DEPI), 2026
- Developing Back-End Apps with Node.js and Express — Coursera | IBM, 2026
- JavaScript Programming Essentials — Coursera | IBM, 2025
- Developing Front-End Apps with React — Coursera | IBM, 2025
- Getting Started with Git and GitHub — Coursera | IBM, 2025

## Additional Information

**Languages:** Arabic – Native · English – C1 (Advanced)